

COM3023

UNIVERSITY OF EXETER

**COLLEGE OF ENGINEERING, MATHEMATICS
AND PHYSICAL SCIENCES**

COMPUTER SCIENCE

Examination, May 2022

Machine Learning and AI

Module Leader: Tinkle Chugh

Duration: TWO HOURS

NO WORD COUNT SPECIFIED

Answer ALL questions.

The marks for this module are calculated from 70% of the percentage mark for this paper plus 30% of the percentage mark for associated coursework.

This is a CLOSED BOOK examination.

Question 1

- (a) What is a conjugate prior in Bayesian inference? State the advantage of using a conjugate prior.

(4 marks)

- (b) State two advantages and two disadvantages of using Bayesian inference.

(6 marks)

- (c) The likelihood of a data set $\mathbf{y} = (y_1, \dots, y_n)$ is described with a Poisson distribution: $p(\mathbf{y}|\lambda) \propto \prod_{i=1}^n \lambda^{y_i} e^{-\lambda}$. Assign a Gamma distribution as the prior on λ : $p(\lambda) \propto \lambda^{\alpha-1} e^{-\beta\lambda}$. In a Gamma distribution the mean is $\frac{\alpha}{\beta}$ and variance is $\frac{\alpha}{\beta^2}$.

- (i) Show that the Gamma distribution on λ is the conjugate prior.

(4 marks)

- (ii) Imagine you have only one data point $y = 10$, What is the mean and variance of the posterior distribution, $p(\lambda|\mathbf{y})$ for $\lambda = 5, \alpha = 25, \beta = 3$?

(4 marks)

(Total 18 marks)

Question 2

- (a) What is the main use of the Laplace approximation in Bayesian statistics?
State one limitation of using the Laplace approximation.

(4 marks)

- (b) The distribution of a random variable θ is given by: $p(\theta|y) = \theta^y(1 - \theta)^{n-y}$.
Estimate the mean and variance using the Laplace approximation. Show your approach in estimating the mean and variance.

(7 marks)

- (c) What is measured by the Kullback–Leibler divergence between two distributions?

(3 marks)

- (d) For the two normal distributions shown in Figure 1, which distribution has the larger entropy and why?

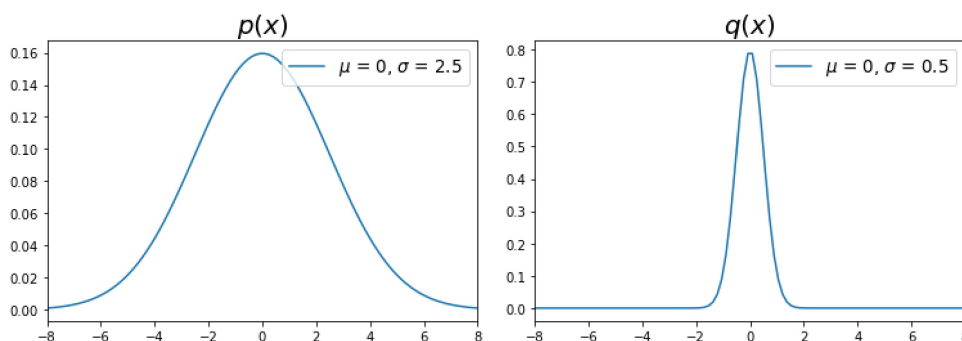


Figure 1: Two univariate normal distributions. The distribution on the left, $p(x)$ has the mean of 0 and standard deviation of 2.5. The distribution on the right, $q(x)$ has the mean of 0 and standard deviation of 0.5

(4 marks)**(Total 18 marks)**

Question 3

(a) What is a Markov process?

(4 marks)

(b) What is hidden in hidden Markov models?

(2 marks)

(c) What are three main problems of interest in hidden Markov models?

(6 marks)

(d) Name two methods for estimating the likelihood of all possible state sequences in hidden Markov models.

(4 marks)

(e) What is the transition probability matrix for the hidden Markov model shown in Figure 2?

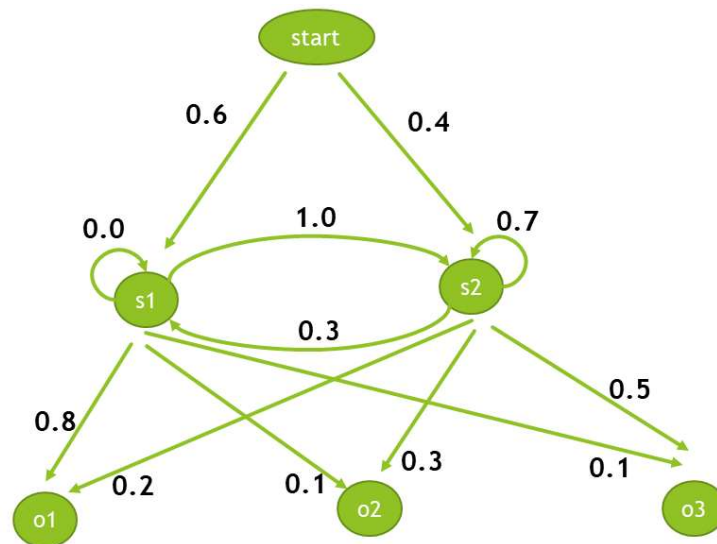


Figure 2: A hidden Markov model. s1 and s2 are states and o1, o2 and o3 are observations.

(4 marks)



Figure 3: A hidden Markov model. 'Rainy', 'Sunny' are states and 'Shopping', 'Cleaning' are observations.

(5 marks)

- (f) In the hidden Markov model shown in Figure 3, the observation sequence is (Shopping $\rightarrow$ Cleaning). How many state sequences are possible? State all possible state sequences.

(Total 25 marks)

Question 4

- (a) Consider a k -armed bandit problem with $k = 4$ actions denoted as 1,2,3 and 4. The initial estimates of values, Q_0 , for all actions is zero, i.e., $Q_0(a) = 0$ for all a . The initial sequence of actions and rewards are given in the following table:

t	Action	Reward
1	3	1
2	1	1
3	2	2
4	2	2
5	1	3

Table 1: Initial sequence of actions and rewards.

- (i) Estimate $Q_i(a)$ where $i = 1, 2, 3, 4, 5$ for all actions using the sample-average technique (1 mark for each correct answer).

(5 marks)

- (ii) What are the optimal actions at each time step?

(5 marks)

- (iii) An ϵ -greedy action selection was used. On some of the time steps the ϵ -greedy selection did not select the greedy action. On which time steps did this definitely occur? Justify your answer.

(7 marks)**(Total 17 marks)**

Question 5

- (a) Find the union and intersection of the following two fuzzy sets A and B :

$$A = \{(x_1, 0.3), (x_2, 0.5), (x_3, 1)\}$$

$$B = \{(x_1, 0.4), (x_2, 0.1), (x_3, 0.9)\}.$$

(4 marks)

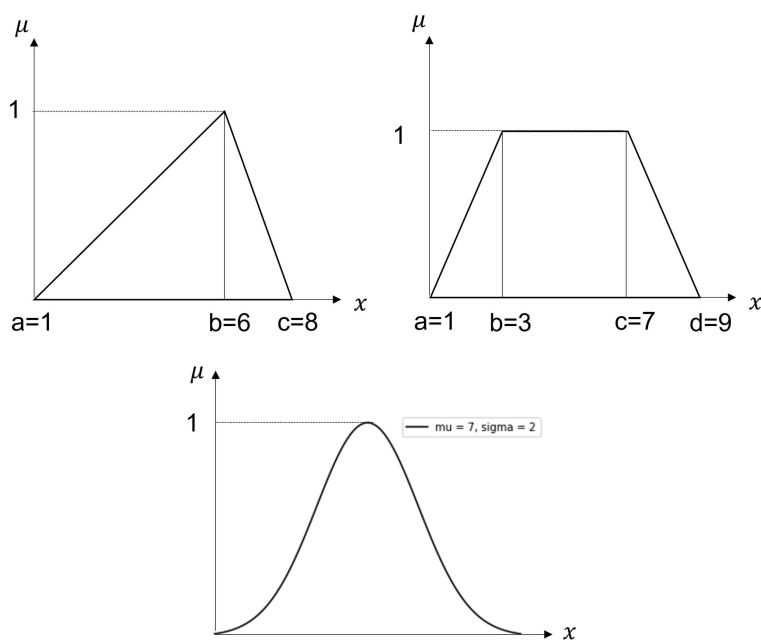
- (b) What is the role of a membership function in a fuzzy set?

(3 marks)

- (c) What are the three main steps in Mamdani fuzzy inference? Provide a short explanation of each step.

(9 marks)

- (d) Estimate the values of the triangular, trapezoidal and Gaussian (with mean of 7 and standard deviation of 2) membership functions shown in Figure 5(d) when $x = 5$.



(6 marks)

(Total 22 marks)